daa6.py

```
n = 8
board = [-1] * n
board[0] = 0

def safe(r, c):
    for i in range(r):
        if board[i] == c or abs(board[i] - c) == abs(i - r):
            return False
    return True

def solve(r):
    if r == n:
        return True
    for c in range(n):
        if safe(r, c):
            board[r] = c
            if solve(r + 1):
                return True
            board[r] = -1
    return False

if solve(1):
    for i in range(n):
        print(" ".join("Q" if board[i] == j else "." for j in range(n)))
else:
    print("No solution exists")

```